

Sai Karthik Kagolanu

Sheffield, UK | 07775 439150 | sai.kagolanu@yahoo.com | Graduate visa to Feb 2027

GitHub | LinkedIn | Portfolio

SUMMARY

MSc Robotics graduate (Sheffield) with hands-on experience building AI agents, ML pipelines and production backends. At Fuse Energy improved billing agent containment from 41% to 74% and reduced unnecessary field visits by 23%. Two hackathon wins. Graduate visa to Feb 2027.

SKILLS

Languages: Python, C++, SQL, JavaScript/TypeScript

ML and data: PyTorch, TensorFlow, scikit-learn, XGBoost, MLflow, LangChain, pandas, OpenCV

Backend and infrastructure: FastAPI, PostgreSQL, Redis, Kafka, Docker, dbt, Dagster, Prometheus/Grafana, GitHub Actions

Other: React/Next.js, Supabase, Cloudflare R2, BigQuery, ClickHouse, AWS, Git

EXPERIENCE

Operations Intern

Fuse Energy

Mar 2026 – Jun 2026

London, UK

- Scaled the billing agent's test suite from 63 to 103 cases covering tool-call leakage and context drift, raising autonomous containment from **41% to 74%** live in production.
- Shipped a second AI assistant for support staff that pulled answers directly from internal docs, removing the Notion pages agents had to search mid-call.
- Deployed ten SQL queries and two Metabase dashboards over ClickHouse to surface metering and install faults early, cutting unnecessary field visits by **23%**.
- Resolved second-line escalations, wrote case files for ombudsman disputes, traced recurring issues to two root causes and rewrote both workflows.

Machine Learning Intern

Infyz Solutions

Jun 2023 – Aug 2023

Hyderabad, India

- Rewrote data preprocessing across three production classifiers in pandas and NumPy, cutting training time by **40%**.
- Benchmarked Random Forest, SVM and XGBoost on 5,000+ labelled samples and recommended the model that shipped, reaching **92% accuracy** in production.

PROJECTS

Zaxia | Code

Jan 2026 – Present

- Delivered a multi-tenant, event-driven backend for bookings, dispatch and payments (FastAPI, PostgreSQL/PostGIS, Redis, Kafka) with 148 passing tests and CI/CD via GitHub Actions.
- Implemented a Virtual COO agent using LangChain and a local LLM with tool-calling over live tenant data including revenue summaries, demand forecasts and booking stats, with graceful degradation when the model is unavailable.
- Ran a full security audit fixing tenant data isolation, replacing an in-memory JWT blacklist with Redis-backed revocation, and adding database-level booking-overlap constraints via PostGIS.

Industrial Intelligence Platform | Code

Jun 2026

- Combined a Digital Twin simulation layer with a Predictive Maintenance pipeline into one system across 10 Docker services, with async telemetry feeding XGBoost failure classification through a shared file interface.
- Orchestrated five Dagster assets on a daily schedule with MLflow experiment tracking and automatic model promotion gated on F1 exceeding 0.70.
- Added a full observability stack (Prometheus, Loki, Grafana) and a BigQuery target switchable via a single config flag, with dbt handling feature engineering across three SQL model layers.

Apartment Society Finance App | Code | Live

Jun 2026 – Present

- Replaced a multi-year Excel workbook with a Next.js and Supabase app covering maintenance billing, levies, PDF receipts and an audit trail enforced by database-level triggers. Live and in daily use by a non-technical treasurer.
- Integrated a read-only Groq AI agent with four query tools and a 10-message history trim so the treasurer can check dues and expenses in plain language without leaving the app.
- Shipped a two-role resident portal and a five-job parallel CI pipeline with automated secret scanning and nightly Vercel Cron backups.

RAG Demo | Code | Live

Jan – Feb 2026

- Developed a hybrid retrieval system over friction stir welding defect documents, combining keyword and semantic search with a re-ranking step for **30% higher accuracy** than keyword-only, with answers returned in under four seconds.
- Released on Streamlit Cloud with conversation memory, query caching and **96%** test coverage.

Computer Vision Classification Suite | Code

Nov 2025 – Mar 2026

- Trained an image classifier on 14,034 photos to **88.9% accuracy**, ported inference to C++ at ~150 ms per image, and surfaced results in a React dashboard with **93%** test coverage.

EDUCATION

MSc Robotics, Merit
University of Sheffield

Sep 2023 – Sep 2024
Sheffield, UK

Dissertation: gesture and speech-controlled robotic arm with **96.98%** gesture accuracy and **95.03%** speech accuracy using custom CNN architectures. Modules: Machine Vision, Machine Intelligence, Deep Learning.

BTech Mechanical Engineering, 8.84/10 (equiv. First Class)
Vellore Institute of Technology

Jul 2019 – May 2023
Chennai, India

Dissertation: Surgical Collaborative Robotic Arm.

AWARDS

Best Use of AI Agents on Arm – HackSheffield10 (2025). Five-agent shopping assistant over live Frasers Group stock, containerised on Arm hardware inside 24 hours.

Winner – Sheffield AI Hackathon (May 2025). *SheffAware*: livability scores for Sheffield's 28 wards using open council data. Live at sheffaware.streamlit.app.